

Ushtrime ND5c

1.3.34 Guri i rëndë $Q=20\text{ N}$ bie nga lartësia $h=5\text{ m}$ dhe futet $\Delta l=5\text{ cm}$ në tokë të butë. Sa është forca mesatare e goditjes në tokë?

Zgjidhja

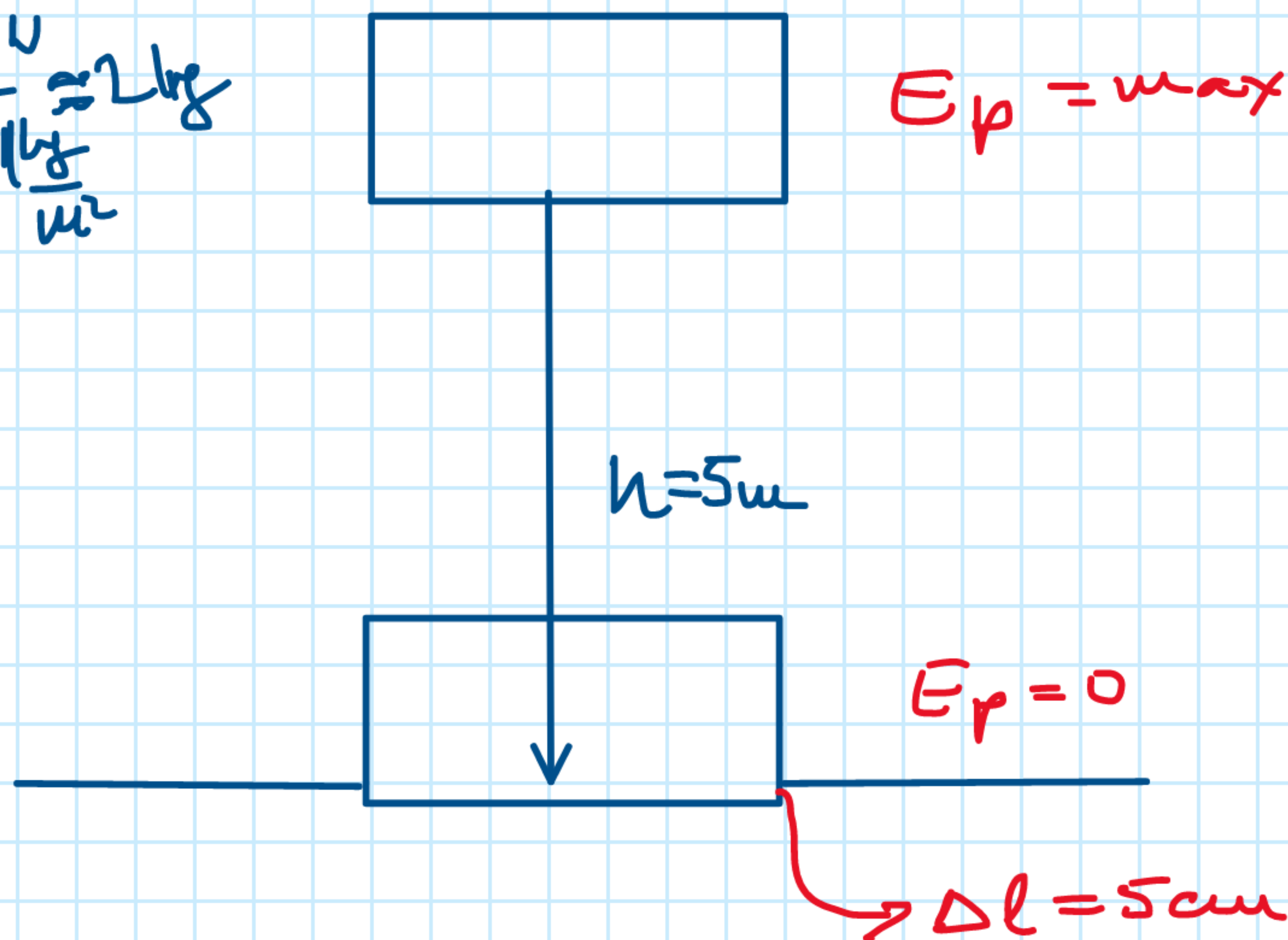
Të dhënat:

$$Q=20\text{ N} \Rightarrow m = \frac{20\text{ N}}{9.81 \frac{\text{m}}{\text{s}^2}} \approx 2\text{ kg}$$

$$h=5\text{ m}$$

$$\Delta l=5\text{ cm}=5 \cdot 10^{-2}\text{ m}$$

$$F=?$$



$$E_p = mgh; \quad F \cdot \Delta l = A$$

$$E_p = A$$

$$mgh = F \cdot \Delta l \Rightarrow F = \frac{Q \cdot h}{\Delta l} = \frac{20\text{ N} \cdot 5\text{ m}}{5 \cdot 10^{-2}\text{ m}} = 20 \cdot 10^2\text{ N}$$

$$F = 2 \cdot 10^3\text{ N} = \underline{\underline{2\text{ kN}}}$$

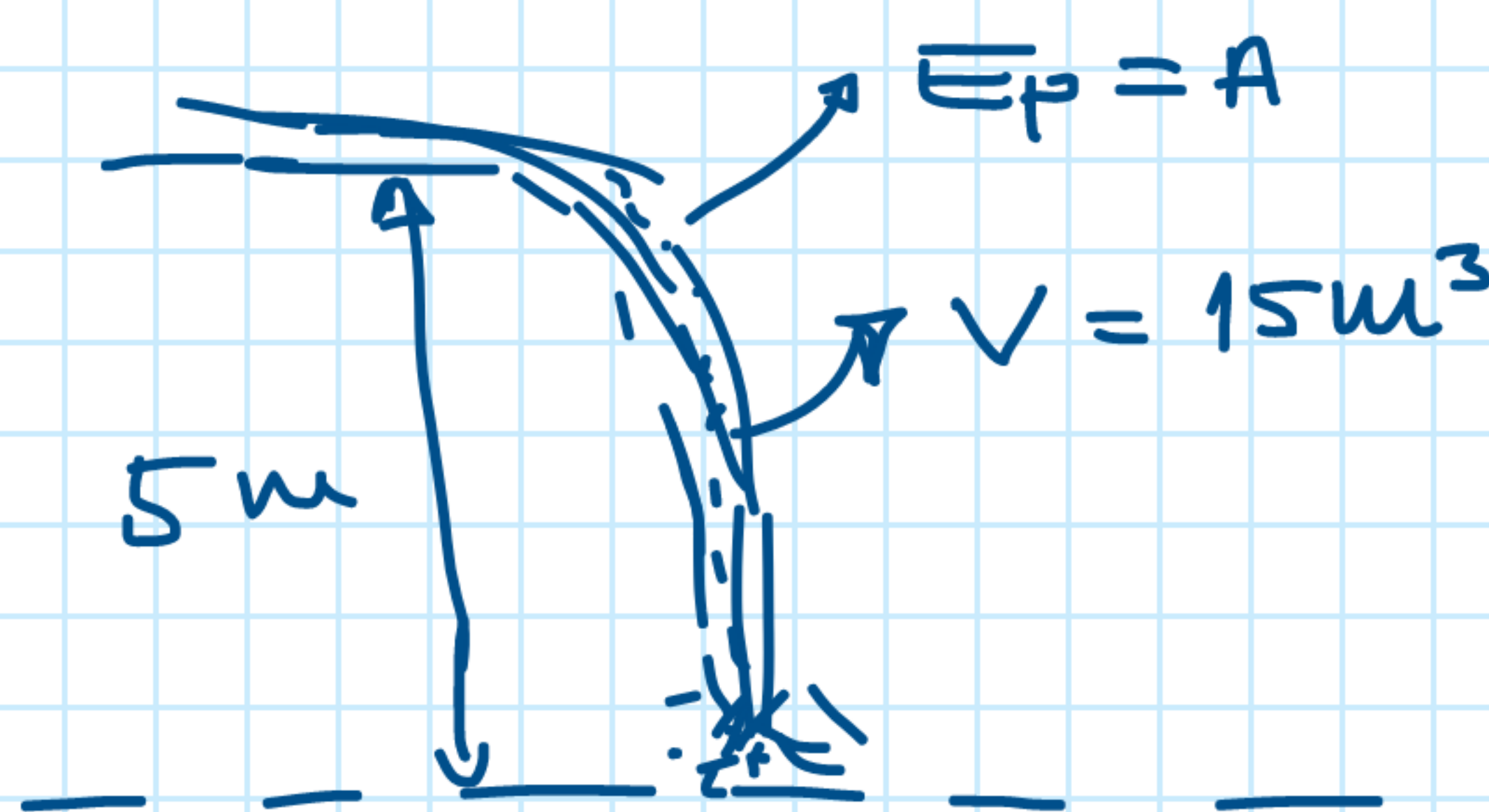
1.3.35 Ujëvara e lartë $h=5\text{ m}$ në çdo sekondë jep $V=15\text{ m}^3$ ujë. Ç'fuqi ka ujëvara?

Zgjidhja

Të dhënat:

$$h=5\text{ m}, \quad g=9.81 \frac{\text{m}}{\text{s}^2}$$

$$V=15\text{ m}^3 (\text{H}_2\text{O})$$



$$P=?$$

$$\rho = 1000 \frac{\text{kg}}{\text{m}^3} (t=4^\circ\text{C})$$

$$P = \frac{A}{t} = \frac{E_p}{t} = \frac{mgh}{t} = \frac{V \cdot \rho \cdot g \cdot h}{t} = \frac{15\text{ m}^3 \cdot 1000 \frac{\text{kg}}{\text{m}^3} \cdot 9.81 \frac{\text{m}}{\text{s}^2} \cdot 5\text{ m}}{1\text{ s}}$$

$$A = F \cdot s = mg \cdot s$$

$$P = 735750\text{ W} = 735.75 \cdot 10^3\text{ W}$$

$$\underline{\underline{P = 735.75\text{ kW}}}$$

$$\frac{m}{V} = \rho \Rightarrow \boxed{m = V \cdot \rho}$$

$$\text{kg} \frac{\text{m}^3}{\text{s}^2} = \frac{\text{kg} \frac{\text{m}}{\text{s}^2} \text{ m}}{\text{s}} = \frac{\text{N} \cdot \text{m}}{\text{s}} = \frac{\text{J}}{\text{s}} = \underline{\underline{\text{W}}}$$